FPGA-Based TCD Headset for Emboli Detection

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Project Summary

Transcranial Doppler (TCD) can identify stroke-causing emboli in the brain, but current tech requires a trained operator, making widespread use difficult.

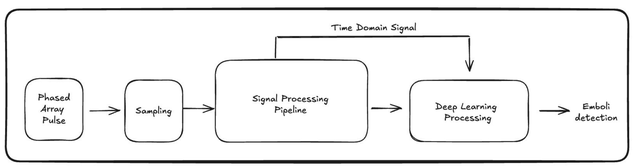
We are building an FPGA-based TCD system which automates embolus detection and stroke risk assessment.

We designed a signal processing pipeline and a simulation framework to validate it while we wait for patient data.

Final System Overview

Final product will use a custom 16-element phased array (designed by our collaborators at Weill Cornell Medicine) to emit ultrasound pulses at 7.5 kHz and listen for echoes from brain tissue.

The pipeline will process raw signals before they are fed into a deep learning model. Both are embedded on an FPGA for computational and energy efficiency.

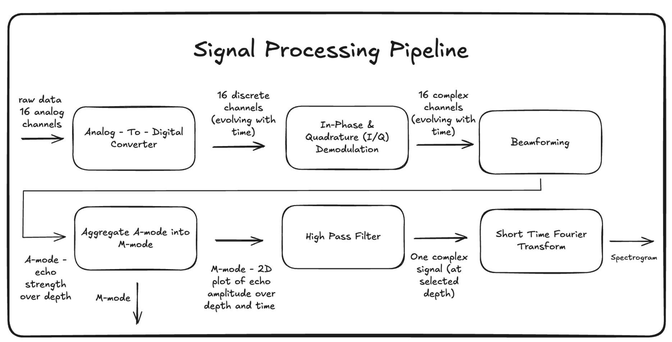


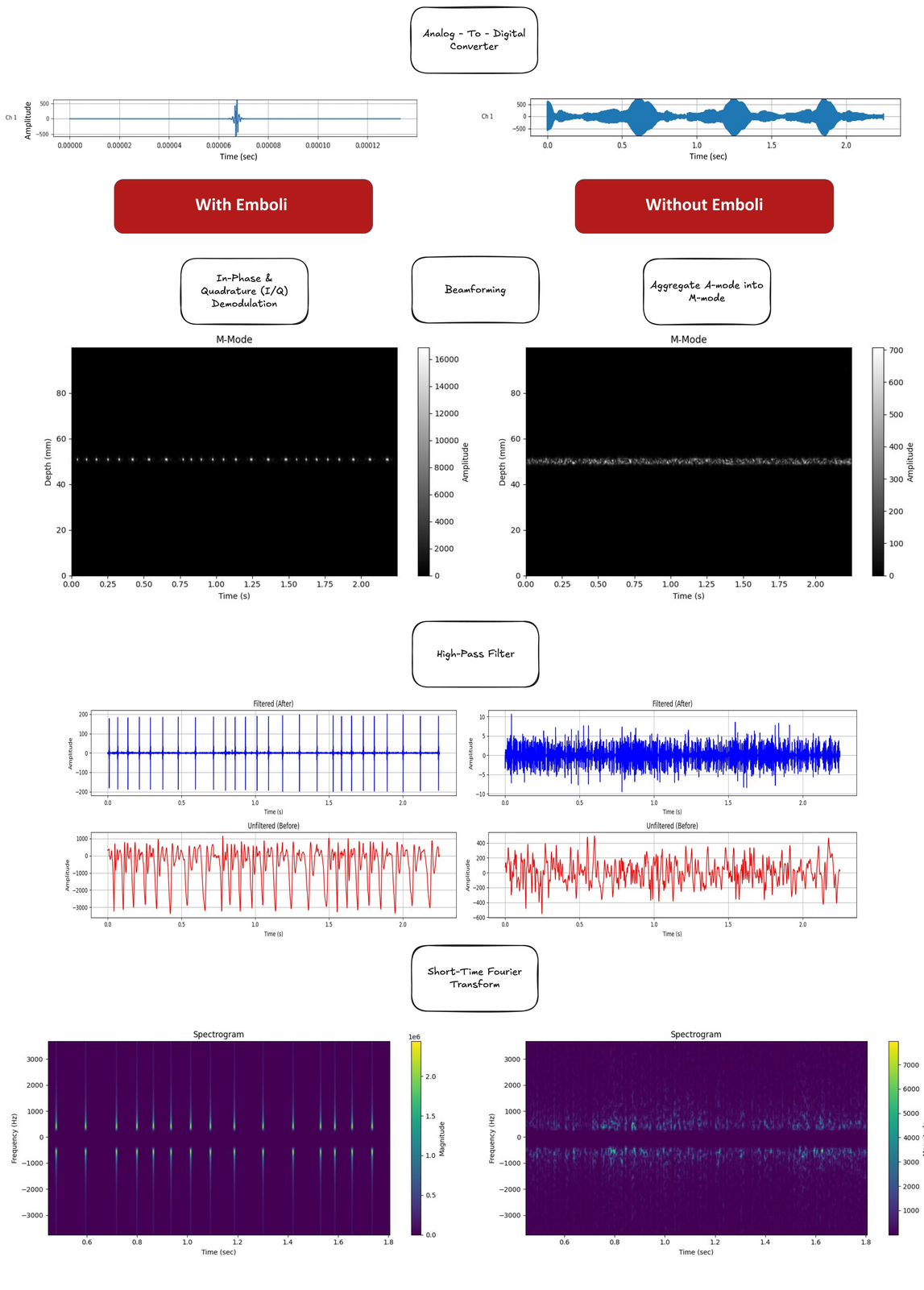
Signal Processing Pipeline

Raw echoes from the phased array are not usable by the model, needs cleaning. We implemented the full pipeline in Python using PyMUST (Matlab Ultrasound Toolbox).

Steps:

- ADC: sample 16 analog channels at 6 MHz (one pulse at a time)
- I/Q demodulation: remove effects of carrier wave by mixing signals with sinusoids and low-pass filtering
- Beamforming: use delay-and-sum to map fast-time samples into 3D space, forming A-mode traces (intensity vs depth)
- Aggregate A-modes over time into M-mode (intensity vs depth x time), M-mode can also be used as input to model
- High-pass filter: remove low frequency cardiac noise and vessel motion
- STFT: generate spectrograms for model input

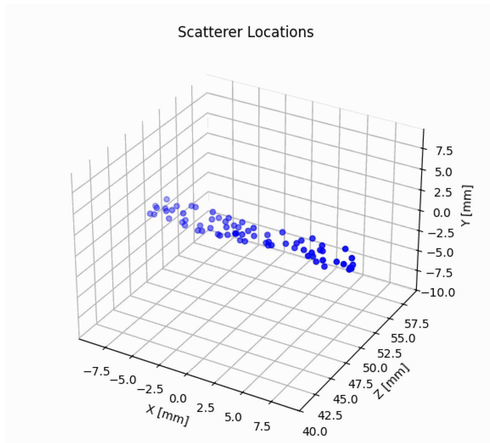


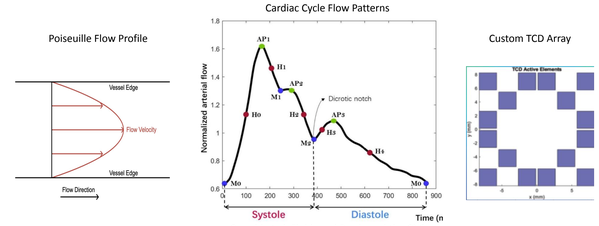


Simulations

Need data to verify processing pipeline, so we built simulation framework to use while waiting for *in vivo* patient data.

Each simulation builds cylindrical blood vessel with scatterers moving according to Poiseuille (laminar) flow. Simulation also mimics how max flow velocity changes over cardiac cycle.





Owashi et al. Fluids and Barriers of the CNS https://doi.org/10.1186/s12987-023-00467-8

Next Steps

Embed pipeline into FPGA fabric, use simulation data to ensure embedded system achieves same results as Python implementation.

Use *in vivo* data collected using phased array by collaborators to train deep learning model and embed it in FPGA fabric.

Acknowledgements

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